

One-Sample T-Test on Sales Data

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The one-sample, two-tailed t-test compares the sample's mean sale value to the population mean of \$390. In this sample of 25 sales records, the average sale is \$407.20. Because this is a sample, we expect the mean to vary somewhat from the population mean due to random sampling error.

The t-test result shows the difference between the sample mean and \$390 is not statistically significant at the 0.05 level ($p \approx 0.203$). Since the p-value is greater than 0.05, we fail to reject the null hypothesis that the true population mean equals \$390. In practical terms, the sample mean is a little higher than \$390, but the evidence is not strong enough to conclude that the true population average sale differs from \$390.

The histogram and Q-Q plot help assess whether the sale values are reasonably normal for using a t-test. The distribution appears generally unimodal, though there may be mild skew and at least one unusually low value compared to the rest of the data. The box plot supports this by showing a potential low outlier (a sale around the mid-250s). Even with a mild outlier, the normality visuals still suggest the data are close enough to normal for a one-sample t-test with $n=25$. Also, the confidence interval for the mean includes \$390 (approximately \$380.07 to \$434.33), reinforcing that \$390 is a plausible population mean.

Overall, the t-test suggests this sample is reasonably consistent with the larger population average of \$390. While the sample mean is higher, the difference is not large enough (relative to the variability in sales) to conclude the sample is unrepresentative of the larger population.

FIGURES AND SCREENSHOTS BELOW

SAS® Studio

▼ Server Files and Folders

odaws01-usw2-2

Folder Shortcuts

Files (Home)

MIS445

CLASS.csv

FederalBankData.

gdp_graph.png

gdp_raw.sas7bda

MIS445FRED-real

sample_pricedata

sasuser.v94

Program 1

CODE LOG

1

The CONTENTS Procedure

Data Set Name	WORK.IMPORT	Observations	25
Member Type	DATA	Variables	28
Engine	V9	Indexes	0
Created	02/06/2026 22:02:07	Observation Length	224
Last Modified	02/06/2026 22:02:07	Deleted Observations	0
Protection		Compressed	NO
Data Set Type		Sorted	NO
Label			
Data Representation	SOLARIS X86 64. LINUX X86 64. ALPHA TRU64. LINUX IA64		

Engine/Host Dependent Information

Data Set Page Size	131072
Number of Data Set Pages	1
First Data Page	1
Max Obs per Page	584
Obs in First Data Page	25
Number of Data Set Repairs	0
Filename	/saswork/SAS_work78210000556E_odaws01-usw2-2.oda.sas.com/SAS_work0F670000556E_odaws01-usw2-2.oda.sas.com/import.sas7bdat
Release Created	9.0401M8
Host Created	Linux
Inode Number	1207962912
Access Permission	rw-r--r--
Owner Name	u64431280
File Size	256KB
File Size (bytes)	262144

Alphabetic List of Variables and Attributes						
#	Variable	Type	Len	Format	Informat	Label
5	cost	Num	8	BEST.		cost
1	date	Num	8	MONYY7.		date
4	discount	Num	8	BEST.		discount
27	line	Num	8	BEST.		line
3	price	Num	8	BEST.		price
6	price1	Num	8	BEST.		price1
7	price2	Num	8	BEST.		price2
8	price3	Num	8	BEST.		price3
9	price4	Num	8	BEST.		price4
10	price5	Num	8	BEST.		price5
11	price6	Num	8	BEST.		price6
12	price7	Num	8	BEST.		price7
13	price8	Num	8	BEST.		price8
14	price9	Num	8	BEST.		price9
15	price10	Num	8	BEST.		price10
16	price11	Num	8	BEST.		price11
17	price12	Num	8	BEST.		price12
18	price13	Num	8	BEST.		price13
19	price14	Num	8	BEST.		price14
20	price15	Num	8	BEST.		price15
21	price16	Num	8	BEST.		price16
22	price17	Num	8	BEST.		price17
28	product	Num	8	BEST.		product
24	productLine	Char	5	\$5.	\$5.	productLine
25	productName	Char	9	\$9.	\$9.	productName
26	region	Num	8	BEST.		region
23	regionName	Char	7	\$7.	\$7.	regionName
2	sale	Num	8	BEST.		sale

Variable: sale (sale)

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.931239	Pr < W	0.0928
Kolmogorov-Smirnov	D	0.126721	Pr > D	>0.1500
Cramer-von Mises	W-Sq	0.064876	Pr > W-Sq	>0.2500
Anderson-Darling	A-Sq	0.484397	Pr > A-Sq	0.2163

Variable: sale (sale)

N	Mean	Std Dev	Std Err	Minimum	Maximum
25	407.2	65.7356	13.1471	254.0	492.0

Mean	95% CL Mean	Std Dev	95% CL Std Dev
407.2	380.1 434.3	65.7356	51.3282 91.4482

DF	t Value	Pr > t
24	30.97	<.0001

